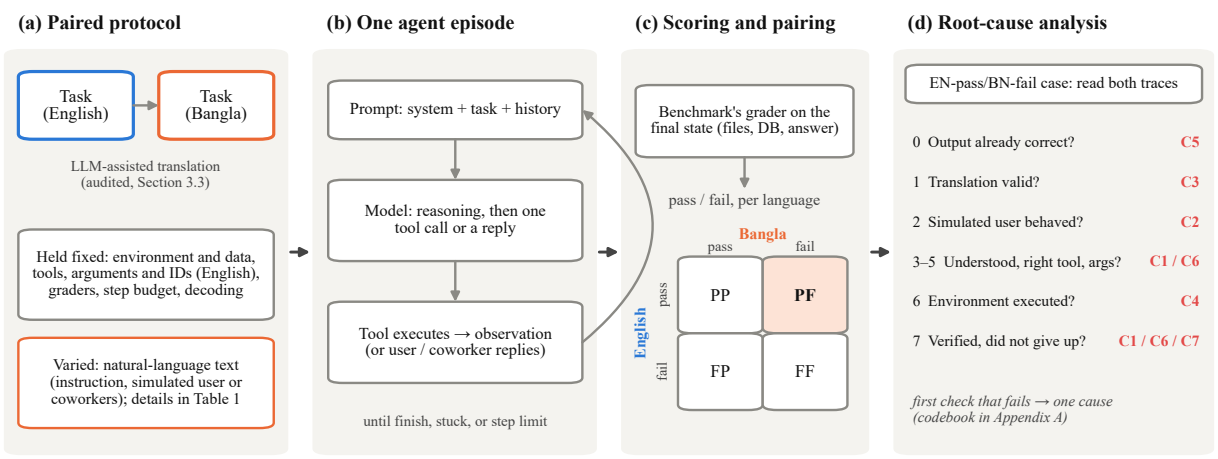

(a) Paired protocol
Task (English)
Task (Bangla)
LLM-assisted translation (audited, Section 3.3)
Held fixed: environment and data, tools, arguments and IDs (English), graders, step budget, decoding
Varied: natural-language text (instruction, simulated user or coworkers); details in Table 1
(b) One agent episode
Prompt: system + task + history
Model: reasoning, then one tool call or a reply
Tool executes → observation (or user / coworker replies)
until finish, stuck, or step limit
(c) Scoring and pairing
Benchmark's grader on the final state (files, DB, answer)
pass / fail, per language
Bangla
pass
fail
English
pass
fail
PP
PF
FP
FF
(d) Root-cause analysis
EN-pass/BN-fail case: read both traces
0 Output already correct? C5
1 Translation valid? C3
2 Simulated user behaved? C2
3–5 Understood, right tool, args? C1 / C6
6 Environment executed? C4
7 Verified, did not give up? C1 / C6 / C7
first check that fails → one cause
(codebook in Appendix A)